New lower bounds for mixed binary/ternary covering codes

Noah Parsons
Independent researcher*

October 2026

Abstract

Let $K(t, b, 1)$ be the minimum size of a code in $\mathbb{F}_3^t \times \mathbb{F}_2^b$ with covering radius 1. We prove $K(2, 5, 1) \geq 34$, $K(1, 7, 1) \geq 45$ and $K(2, 6, 1) \geq 59$, improving the lower bounds 32, 42 and 57 in the tables of mixed binary/ternary covering codes. These are three of the five smallest open entries for covering radius 1. The proofs follow the codeword-distribution method of Östergård and Blass: the space is divided into cells, a counting inequality restricts the number of codewords in each cell, and every remaining distribution is shown infeasible by integer programming. Each case is decided by two independent solvers, Gurobi and CPLEX, and the steps that produce the cases are checked against the definitions by separately written code.

1 Introduction

A code $C \subseteq \mathbb{F}_3^t \times \mathbb{F}_2^b$ has *covering radius* 1 if every word is within Hamming distance 1 of some codeword. The minimum size of such a code is denoted $K(t, b, 1)$ [2]. Bounds for $t + b \leq 13$ are collected in the tables of Kéri [4], which follow the tables of Östergård and Hämäläinen [7] and of Bertolo, Östergård and Weakley [1]; the tables were last updated in 2011.

Ordered by the size $3^t 2^b$ of the space, the five smallest open entries for covering radius 1 are $K(2, 5, 1)$, $K(1, 7, 1)$, $K(3, 4, 1)$, $K(5, 1, 1)$ and $K(2, 6, 1)$ (spaces of 288, 384, 432, 486 and 576 words). This note improves the lower bound of three of them.

Theorem 1. *$K(2, 5, 1) \geq 34$, $K(1, 7, 1) \geq 45$ and $K(2, 6, 1) \geq 59$.*

Entry	Previous bounds [4]	Sources	New lower bound
$K(2, 5, 1)$	32–36	[5], [3]	34
$K(1, 7, 1)$	42–48	[5], $3K(0, 7, 1)$	45
$K(2, 6, 1)$	57–64	[5], [3]	59

Table 1: The three entries. All three previous lower bounds are due to Lang, Quistorff and Schneider [5]. The upper bounds are due to Hämäläinen and Rankinen [3], and for $K(1, 7, 1)$ to the direct product of $\mathbb{F}_3$ with the binary Hamming code of length 7.

We are not aware of improvements to these entries after 2011. Since a covering code with fewer than M words can be enlarged to exactly M words, a lower bound $K \geq M + 1$ follows from the nonexistence of covering codes with exactly M' words for each M' from the previous lower bound up to M.

*parsons.m.noah@gmail.com, ORCID 0009-0000-7224-6040.

2 Method

2.1 Cells and the counting inequality

Fix a set T of coordinates. For each assignment x of values to the coordinates in T, the *cell* S_x is the set of words that agree with x on T. It has 2^f words when the free coordinates (those outside T) are f binary coordinates, as in all cases below. Two cells are *adjacent* if their assignments differ in exactly one coordinate. For a code C write $M_x = |C \cap S_x|$.

Lemma 2. *If C has covering radius 1 and the free coordinates are f binary coordinates, then for every cell S_x*

$$(1+f)\,M_x + \sum_{y \text{ adjacent to } x} M_y \;\geq\; 2^f.$$

Proof. The balls of radius 1 around the codewords cover the 2^f words of S_x. A codeword in S_x covers itself and the f words of S_x obtained by changing one free coordinate. A codeword in an adjacent cell differs from S_x in exactly one fixed coordinate, so its ball meets S_x in exactly one word. A codeword in any other cell differs from every word of S_x in at least two coordinates, so its ball misses S_x. □

Lemma 2 is the inequality of Östergård and Blass [6], used there for $K(9,1) = 62$, written for mixed alphabets. We use two kinds of cells.

Two ternary coordinates. For $K(2,b,1)$ we fix both ternary coordinates. The nine cells form a 3×3 grid, each cell has $f = b$ free coordinates, and a cell is adjacent to the four cells in its row and column. Writing R and C for the totals of the row and column of a cell, the inequality becomes

$$(b-1)\,M_x + R + C \;\geq\; 2^b, \tag{1}$$

since M_x is counted once in R and once in C.

One ternary coordinate. For $K(1,7,1)$ we fix the ternary coordinate and then the binary coordinates $v_1, v_2, \ldots$ one at a time, as in [6]. At level m there are $3 \cdot 2^m$ cells with $f = 7 - m$; a cell (a,x) is adjacent to the two cells (a',x), $a' \neq a$, and to the m cells $(a, x+e_i)$. Each distribution at level $m+1$ refines one at level m, and the inequality is imposed at every level.

2.2 Symmetry and exact cases

Permuting the symbols of a ternary coordinate, permuting coordinates of equal alphabet size, and complementing binary coordinates are isometries, so equivalent distributions need only be considered once. For the grid these maps act as permutations of rows, permutations of columns and transposition, a group of order 72. For the levels of $K(1,7,1)$ at level 2 they act as the 3! permutations of the ternary symbols and the $2^2 \cdot 2!$ signed permutations of v_1, v_2, a group of order 48.

Each remaining distribution becomes an integer program with one binary variable per word, one covering constraint per word (the sum of the variables in its ball is at least 1) and one constraint per cell fixing the number of codewords in it. A covering code with M words exists if and only if at least one of these programs is feasible.

3 Computation

All computations were run single-threaded on one core of an Intel Core Ultra 7 255HX, with Gurobi 13.0 (gurobipy, size-limited licence) and IBM CPLEX 22.2 (Community Edition). The largest models, for $K(2,6,1)$, have 576 variables and 585 constraints, within both licences.

Entry	M	distrib.	cases	Gurobi	CPLEX	further solvers
$K(2,5,1)$	32	306	10	all inf.	all inf.	SCIP, HiGHS: all inf.
	33	1 194	32	all inf.	all inf.	SCIP: all; HiGHS: 31 of 32
$K(1,7,1)$	42	596	24	all inf.	all inf.	SCIP: all inf.
	43	3 936	109	all inf.	all inf.	SCIP: all inf.
	44	18 108	489	all inf.	all inf.	–
$K(2,6,1)$	57	372	14	all inf.	all inf.	–
	58	2 637	64	all inf.	all inf.	–

Table 2: The computation. "Distrib." counts the distributions over the cells (for $K(1,7,1)$, at level 2) that satisfy Lemma 2; "cases" counts them up to symmetry, the integer programs solved; "inf." means infeasible. One HiGHS case for $K(2,5,1)$ was still running after 30 minutes and was not completed; SCIP was stopped after part of the $K(1,7,1)$ cases at $M = 44$ for lack of speed.

Table 2 lists the cases. Every case of every entry was found infeasible by both Gurobi and CPLEX. The longest single case took 54 seconds with Gurobi and 10 minutes with CPLEX (both for $K(2,6,1)$ at $M = 58$); the total for all cases was under 30 minutes with Gurobi and under one hour with CPLEX. SCIP 10.0 and HiGHS (through SciPy 1.16.3) were used as further checks where their speed allowed.

4 Verification

A false "no code exists" could come from a wrong counting inequality, an incomplete list of distributions, a faulty symmetry reduction, a wrong integer program, or a solver error. Each was checked separately.

1. **The inequality from the definition.** For every cell S_x and every word u, the number of words of S_x whose ball contains u was computed from Hamming distances alone. In every case it depends only on the cell of u and equals the coefficient in Lemma 2: $1 + f$ for $u \in S_x$, 1 for adjacent cells and 0 otherwise. This was done for the grid with $b = 5, 6$ and for levels 0, 1, 2 of $K(1,7,1)$.

2. **Completeness of the distributions.** For $K(2,5,1)$, a separate vectorised enumeration of all 76 904 685 ways to place 32 words in nine cells (95 548 245 for 33), using the coefficients computed in step 1, gives exactly the 306 (1 194) distributions of Table 2. For $K(1,7,1)$, a separate brute-force enumeration through levels 0, 1, 2 gives exactly 596, 3 936 and 18 108. For $K(2,6,1)$, where a full enumeration is too slow, the case lists come from a fast enumerator based on (1); on seven smaller instances ($b = 3, 4, 5$) its output is identical to that of the direct brute-force enumerator.

3. **Symmetry reduction.** With the group elements written out by separate code, every distribution in step 2 is equivalent to one of the cases solved, and every case solved occurs.

4. **The integer program.** A real covering code satisfies the program for its own cell distribution, and the same code with one word removed does not: a 36-word code for $K(2,5,1)$, the 48-word record code for $K(1,7,1)$, and a 74-word code for $K(2,6,1)$ (all confirmed by an independent brute-force checker).

5. **Known values.** The same pipelines reproduce $K(2,3,1) = 12$, $K(2,4,1) = 20$, $K(1,5,1) = 16$ and $K(1,6,1) = 24$: each reports no code one below the known value and finds a code at the known value, confirmed by the checker. Gurobi and CPLEX each pass these controls.

6. **Solvers.** Every case was decided by two solvers from different developers. The models for $K(2,5,1)$ and $K(2,6,1)$ were built by separate code for each solver; for $K(1,7,1)$ both solvers received models built by the same code, which step 4 tests.

Remark 3. All the solvers use floating-point arithmetic, so each verdict is evidence rather than an exact certificate. The independent agreement of two commercial-grade solvers on every case, together with the checks above, is the basis of Theorem 1.

5 Remarks

1. For $K(1,7,1)$ we showed separately [8] that any covering code with at most 47 words has trivial automorphism group. With Theorem 1, a code improving the upper bound 48 would have 45, 46 or 47 words and no symmetry.

2. The method continues to the next sizes: $K(2,5,1)$ at $M = 34$ has 93 cases up to symmetry, $K(1,7,1)$ at $M = 45$ has 1 848, and $K(2,6,1)$ at $M = 59$ has 221. These were not completed.

3. The entries $K(3,4,1)$ and $K(5,1,1)$ have three or more ternary coordinates, which the programs used here do not handle; they were not attempted.

Code and data. All programs and logs are in the public repository

`https://github.com/Noah-Parsons/covering-codes`

- grid method: `src/split_t2.py`;
- its rechecks: `src/recheck_t2_gurobi.py`,

  `src/recheck_t2_cplex.py`, `src/recheck_t2_highs.py`;
- its checks: `src/verify_t2_distributions.py`;
- level method: `src/exact_dist.py`, `src/leaves_t1.py`; checks `src/verify_t1_leaves.py`;
- brute-force code checker: `src/checker.py`; logs in `results/`.

References

[1] R. Bertolo, P. R. J. Östergård and W. D. Weakley, An updated table of binary/ternary mixed covering codes, *J. Combin. Des.* 12 (2004), 157–176.

[2] G. Cohen, I. Honkala, S. Litsyn and A. Lobstein, *Covering Codes*, North-Holland, Amsterdam, 1997.

[3] H. Hämäläinen and S. Rankinen, Upper bounds for football pool problems and mixed covering codes, *J. Combin. Theory Ser. A* 56 (1991), 84–95.

[4] G. Kéri, Tables for bounds on covering codes, `https://old.sztaki.hu/~keri/codes/index.htm` (accessed 16 October 2026).

[5] W. Lang, J. Quistorff and E. Schneider, New results on integer programming for codes, *Congr. Numer.* 188 (2007), 97–107.

[6] P. R. J. Östergård and U. Blass, On the size of optimal binary codes of length 9 and covering radius 1, *IEEE Trans. Inform. Theory* 47 (2001), 2556–2557.

[7] P. R. J. Östergård and H. O. Hämäläinen, A new table of binary/ternary mixed covering codes, *Des. Codes Cryptogr.* 11 (1997), 151–178.

[8] N. Parsons, Improving the covering code bound $K(1, 7, 1) \leq 48$ requires an asymmetric code, manuscript, 2026.